

MACHINE LEARNING FOR BUSINESS APPLICATION

ASSIGNMENT – 02

Customer Churn Prediction Using Machine Learning

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SECTION: FBA-3

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1. Introduction

Customer churn is an important business issue for telecom companies because customers who discontinue their services can reduce recurring revenue and increase customer acquisition costs. This project develops a machine learning classification model to predict whether a telecom customer is likely to churn using demographic, account, and service-related information.

The project follows a structured business analytics workflow: defining the business problem, preparing customer data, performing exploratory data analysis, training a classification model, evaluating its performance, generating churn probabilities, and saving the trained model and preprocessing object for future application use.

2. Business Problem

A telecom company may lose customers when they discontinue its services. Customer churn can reduce revenue and increase the cost of acquiring new customers. Therefore, the company needs to identify customers who are likely to leave so that appropriate retention actions can be taken.

Analytics Objective: Develop a classification model that predicts whether a customer is likely to churn based on customer demographic, account, and service-related information.

2.1 Target Variable

Item	Description
Target variable	Churn
Yes	Customer has churned
No	Customer has not churned

Business Decision: The model can help the company identify customers at higher risk of churn. Based on the prediction, the company can take proactive retention actions such as suitable offers, improved customer service, or reviewing customer plans.

3. Dataset Description

The notebook uses the Telco Customer Churn dataset file named "WA_Fn-UseC_-Telco-CustomerChurn.csv". The dataset contains customer-level telecom service, account, and demographic variables, with Churn used as the target variable.

Dataset Attribute	Value / Description
Dataset name	Telco Customer Churn
Source/file used	WA_Fn-UseC_-Telco-Customer-Churn.csv
Rows after cleaning	7,032
Columns after cleaning	20
Target variable	Churn
Target type	Binary classification
Identifier removed	customerID

3.1 Variables

The notebook separates predictors into numerical and categorical variables using their data types. Numerical variables are standardized, while categorical variables are one-hot encoded. The target variable Churn is mapped from Yes/No to 1/0.

The dataset includes customer demographic, account, and service-related information. The analysis focuses on variables such as tenure, contract type, monthly charges, total charges, internet service and other customer/service characteristics.

4. Data Preparation

- Data loading: The dataset was loaded into a Pandas DataFrame using `pd.read_csv()`.
- Missing values, blank values, duplicate records, and data types were checked.
- TotalCharges was converted to numeric using coercion for invalid values.
- Rows containing missing values after the conversion were removed.
- customerID was removed because it is an identifier rather than a predictive business feature.
- The final dataset contained 7,032 rows and 20 columns.
- Numerical variables were standardized using StandardScaler.
- Categorical variables were encoded using OneHotEncoder(`handle_unknown='ignore'`).
- The preprocessing object was fitted on the training data and then applied to the test data to maintain a consistent modelling process.

5. Exploratory Data Analysis

Five visualizations were produced in the notebook to understand customer churn patterns before model training.

Customer Churn Distribution

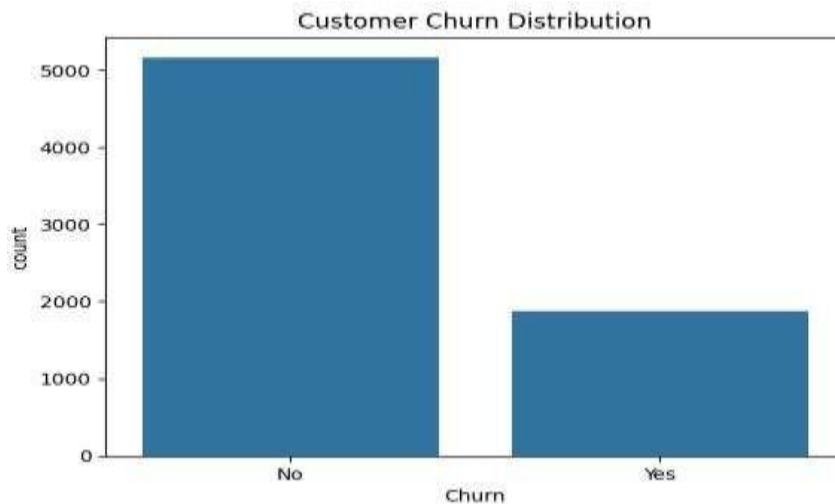


Figure 1: Customer Churn Distribution

Most customers in the dataset did not churn, while a smaller group discontinued the service. This indicates an imbalanced target and supports the need to consider metrics beyond accuracy when evaluating the model.

Customer Churn by Contract Type

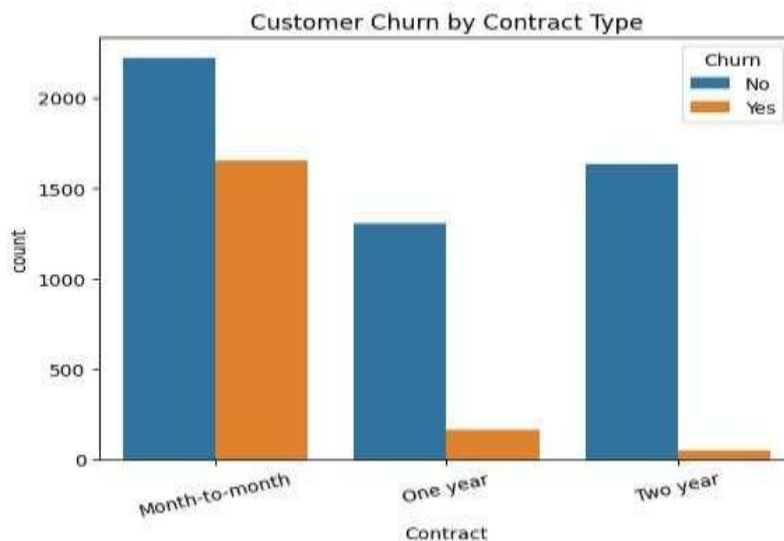


Figure 2: Customer Churn by Contract Type

Churn varies across contract types. Contract structure therefore provides a useful business segmentation variable for identifying customers who may need additional retention attention.

5. Exploratory Data Analysis (continued)

Customer Churn by Tenure

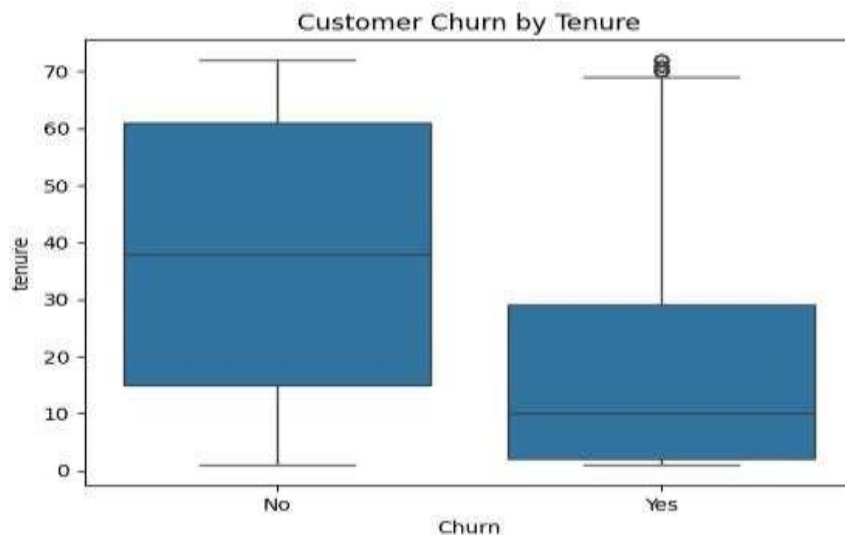


Figure 3: Customer Churn by Tenure

The tenure distributions differ between churned and retained customers. Tenure is therefore a useful signal for identifying customers at different stages of the customer relationship.

Customer Churn by Monthly Charges

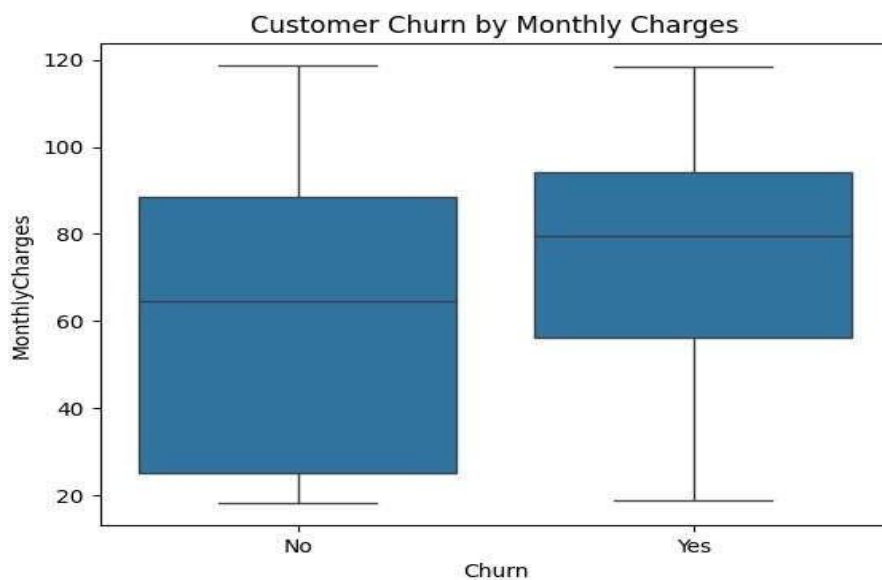


Figure 4: Customer Churn by Monthly Charges

Monthly charges show differences between churn categories. Charge levels can therefore be considered when designing targeted retention strategies, particularly where pricing or plan suitability may be relevant.

Correlation Heatmap

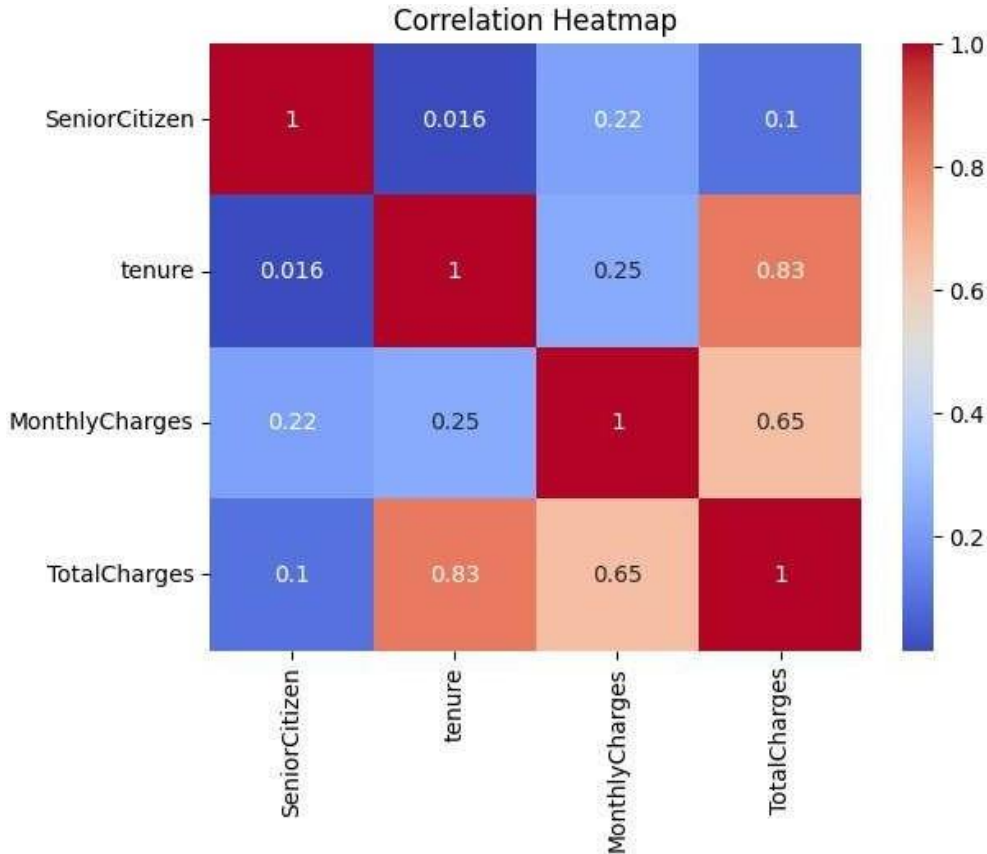


Figure 5: Correlation Heatmap

The heatmap shows relationships among the numerical variables. These relationships provide supporting evidence for feature understanding before model development.

6. Machine Learning Methodology

This is a binary classification problem in which the target variable Churn is encoded as 1 for Yes and 0 for No.

- Feature and target separation: Churn was separated from the predictor variables.
- Train-test split: An 80:20 split was used with `random_state=42` and stratification on the target.
- Numerical preprocessing: `StandardScaler` was applied to numerical variables.
- Categorical preprocessing: `OneHotEncoder(handle_unknown='ignore')` was applied to categorical variables.
- Model: Logistic Regression was trained using the processed training data.
- Prediction: The model generated both class predictions and churn probabilities.

The model and preprocessing object were saved separately using `joblib`. This allows the same preprocessing steps used during training to be reused when the model is applied to new customer information.

6.1 Feature Analysis

Feature	Coefficient	Absolute Importance
num__tenure	-1.352313	1.352313
cat__Contract_Two year	-0.779030	0.779030
num__TotalCharges	0.644014	0.644014
cat__InternetService_DSL	-0.616132	0.616132
cat__Contract_Month-to-month	0.613846	0.613846
cat__InternetService_Fiber optic	0.590184	0.590184
num__MonthlyCharges	-0.541006	0.541006
cat__PaperlessBilling_No	-0.300387	0.300387
cat__MultipleLines_No	-0.293470	0.293470
cat__InternetService_No	-0.283724	0.283724

The largest absolute coefficients recorded in the notebook include tenure, contract type, TotalCharges, InternetService and MonthlyCharges. Because the coefficients are based on transformed variables, they should be interpreted as model signals rather than causal effects.

7. Model Evaluation

The Logistic Regression model was evaluated on the held-out test dataset using Accuracy, Precision, Recall and F1-score.

Metric	Result
Accuracy	80.38%
Precision	64.85%
Recall	57.22%
F1 Score	60.80%

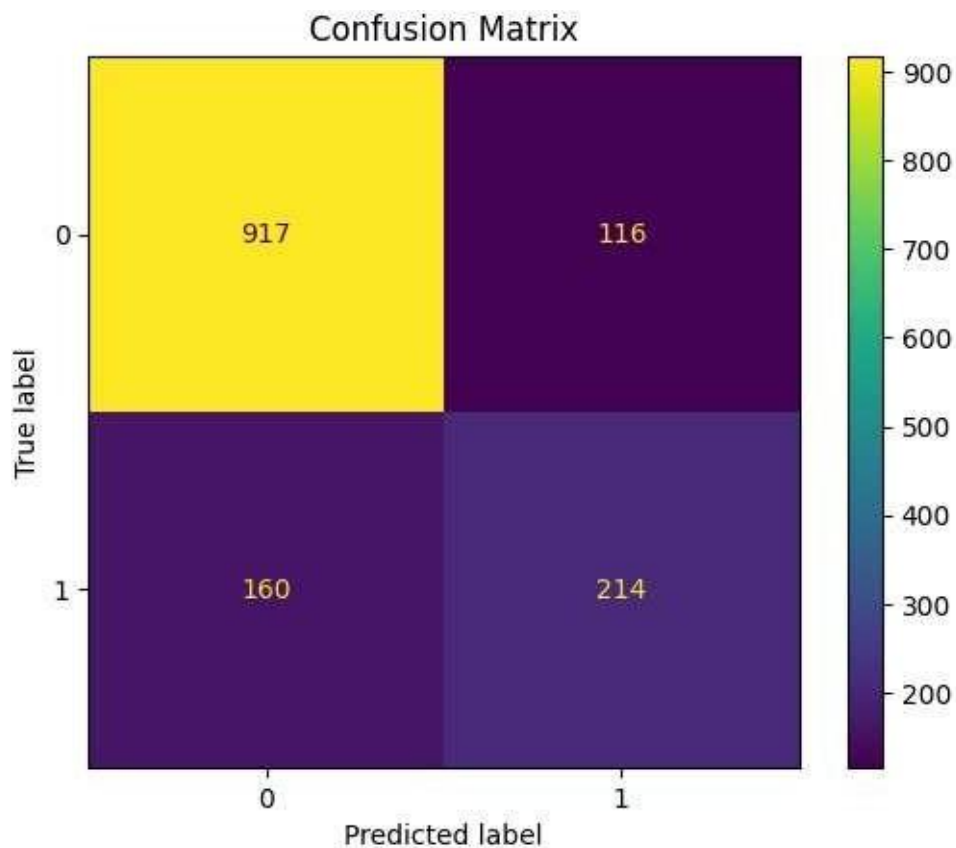


Figure 6: Confusion Matrix

The model correctly identifies a substantial proportion of retained and churned customers, but it also produces both false-positive and false-negative predictions. The recall of 57.22% indicates that some customers who actually churn are not identified by the model.

For the business, a false negative may mean that a genuine churn-risk customer does not receive a retention intervention. A false positive may result in retention resources being spent on a customer who would otherwise remain. Therefore, the model should be treated as decision support rather than as a guarantee of churn.

8. Streamlit Application

The supplied notebook trains the Logistic Regression model, generates churn probabilities, and saves the model and preprocessing object using joblib. A complete Streamlit interface is not present in the supplied notebook.

For an application stage, the saved preprocessing object should transform new customer inputs in the same way as the training data, after which the saved Logistic Regression model can return a churn prediction and probability. The prediction can then be presented as a business risk indicator for retention review.

9. Deployment

The supplied notebook does not provide a live Streamlit deployment URL or deployment configuration. Therefore, no live application link is claimed in this report. The saved model and preprocessing files provide the technical artifacts required for a later application or deployment stage.

10. Business Insights

- Most customers are retained, but the smaller churn group represents an important opportunity for proactive retention.
- Contract type is associated with different churn patterns, so contract structure can be used for customer segmentation and retention planning.
- Tenure is an important model signal; customers at different stages of the relationship can require different retention approaches.
- Monthly charges and TotalCharges provide useful signals for reviewing pricing, plan suitability and customer value.
- The model's churn probability can be used to rank customers for retention review instead of treating all customers equally.

11. Limitations

- Model performance limitation: The model achieved 80.38% accuracy, 64.85% precision, 57.22% recall and 60.80% F1-score. The recall indicates that some actual churners are still missed.
- Data limitation: The analysis is based on the variables available in the Telco Customer Churn dataset and does not include every factor that may influence churn.
- Feature limitation: Customer satisfaction, complaints, competitor offers and detailed service experience are not represented in the model, so some important churn drivers may be absent.
- Model limitation: Logistic Regression is relatively simple and may not capture complex or nonlinear relationships between customer characteristics and churn.
- Application limitation: Any future Streamlit interface depends on valid user inputs and consistent preprocessing. Model performance may also change when customer behaviour or business conditions change.
- Decision-risk limitation: False negatives can cause genuine churn-risk customers to be missed, while false positives can lead to retention resources being spent on customers who would not have churned.

12. Recommendations

- Use the model as a screening tool to prioritize customers with a higher predicted probability of churn. High-risk customers should be reviewed by the retention team before an action is selected.
- Target retention actions using contract type, tenure, monthly charges, internet-service characteristics and other relevant customer information instead of applying the same offer to every customer.
- Combine churn probability with customer value and service history so that retention resources are focused on customers where intervention is most relevant.
- Review false-positive and false-negative cases regularly to identify where the model is making mistakes and whether changes to the data, features or modelling approach are required.

- Re-evaluate and retrain the model as new customer data becomes available so that its predictions continue to reflect current customer behaviour and business conditions.

13. Conclusion

This project developed a machine-learning based customer churn prediction solution using the Telco Customer Churn dataset. The workflow covered data cleaning, exploratory data analysis, preprocessing of numerical and categorical variables, Logistic Regression model training, model evaluation, probability generation and saving of the trained model and preprocessing object.

On the test dataset, the model achieved 80.38% accuracy, 64.85% precision, 57.22% recall and a 60.80% F1-score. The model also generates churn probabilities, which can help the business prioritize customers according to their estimated risk rather than treating all customers equally.

Feature analysis highlighted tenure, contract type, TotalCharges, InternetService and MonthlyCharges among the variables with comparatively larger absolute coefficients. These provide useful areas for further retention analysis, while the coefficients should not be interpreted as proof of causation.

Overall, the project demonstrates how customer data can be transformed into a churn-risk prediction and used as decision support for retention planning. The predictions should be combined with managerial judgement and additional customer information before retention action is taken.

14. GitHub & Deployment Links

- GitHub Repository: <https-github.com-logeshwar25002961-lang-Customer-Churn-Prediction-MLBA>
- Live Application: Customer Churn Prediction – <https://customer-churn-prediction-mlba-gadk3xlgzudwnsi6ggadts.streamlit.app/>